

Tinkering & Mentoring - VACC 201

Problem Identification, Ideation & Initial Design



ParkSmart: Intelligent Vehicle Management System

Date : 5/05/26

Name of the Presenter: Grishma, Hanee, Prit,
Krutika and Krishi

Email Id:

Mobile No.:

Faculty Mentor Name: Dr. Ankit Sudhir

Team No.: 4

Semester: 2nd

Program: Bsc. Biotechnology

School: School of Science

Student Team

Name	Enrollment No.	Key Responsibility
Grishma Gurjar	25sc02019	Implementation Planning
Hanee Suthar	25sc02020	Research & Planning
Prit Kanojiya	25sc02022	UI design
Krutika Karnet	25sc02023	Data handling
Krishi Kayastha	25sc02024	Documentation & PPT

Faculty Mentor

Name	Designation	Department, School
Dr. Ankit Sudhir		SOS_04

Brief of Tinkering Project

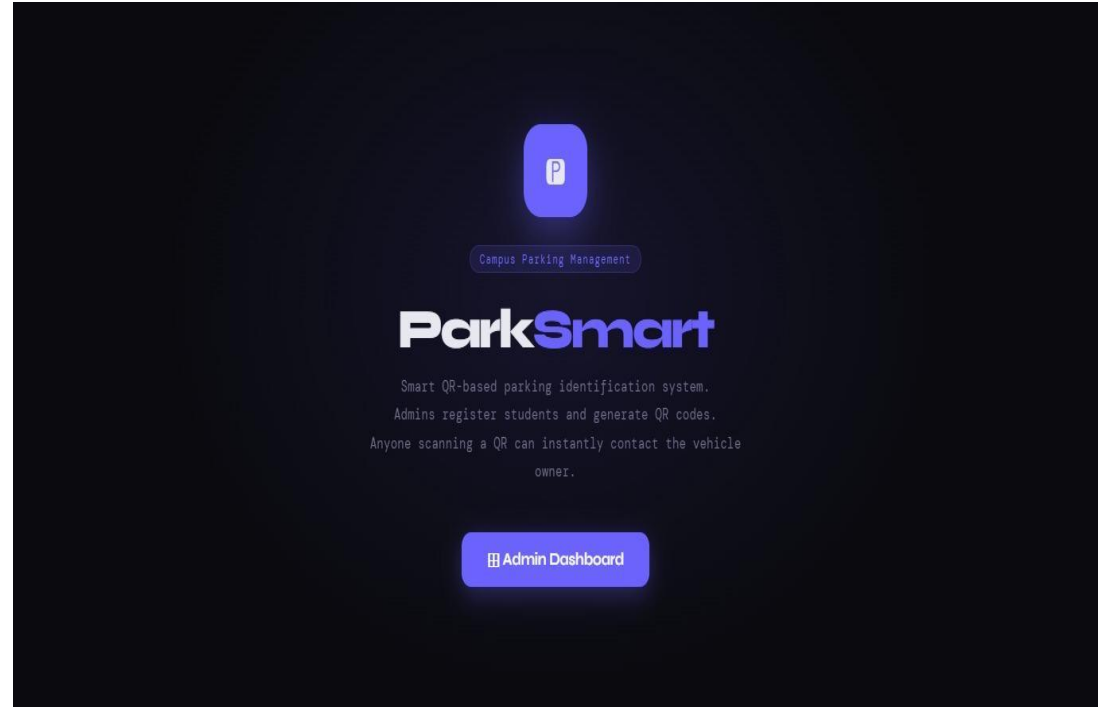
- **Title of Tinkering Project:** ParkSmart: Intelligent Vehicle Management System
- **Sector:** Information Technology (Smart Parking Smart)
- **Brief of Tinkering Project:** A Smart QR-Based Vehicle Identification System is proposed for efficient campus management.
- Each vehicle is assigned a barcode containing student details, enabling quick, identification and communication in case of parking issues.
- The system can also support entry-exit tracking and attendance, improving security and automation.

PROBLEM STATEMENT

- Parking on campus is messy and unorganised
- Hard to find vehicle owners quickly
- No proper tracking of vehicles
- Manual checking wastes time.

OUR SOLUTION

- QR code-based vehicle identification system
- Each vehicle gets a unique QR code
- Scan→ Gets student details instantly
- Easy communication with owner.



HOW IT WORKS

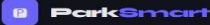
1. Student registers vehicle
2. QR is generated
3. Sticker placed on vehicle
4. Scan using phone
5. Details appear instantly

OBJECTIVE

USP/ Novelty/ Unique Points of Tinkering Project

- Low cost (no expensive tech)
- Works on any smart device
- Instant identification
- Combines parking + security + tracking
- Scalable for other campuses

System Interference



MENU

Dashboard

Students

Reports

Settings

ParkSmart Admin v1.0

System Active

Admin Dashboard

Parking Management System

Admin Mode

TOTAL STUDENTS

4

QR CODES ISSUED

4

PROGRAMS REGISTERED

2

Register Student

FULL NAME

e.g. Raj Sharma

BRANCH

Select Branch

PROGRAM

Select Program

PHONE NUMBER

e.g. 9876543210

VEHICLE NUMBER

E.G. GJ06AB1234

ENROLLMENT / ROLL NO.

e.g. 22CS001

+ Register & Generate QR

Registered Students

Search by name, branch, roll no...

STUDENT	PROGRAM	PHONE	VEHICLE	ACTIONS
hanee suthar Other · 25sec02020	PhD	8154836019	GJ16AJ4873	QR View Del
Khushi Agrawal Computer Science · 25sec02079	B.Tech	6359525281	GJ06FH9824	QR View Del
krutika karnet Other · 25sec02023	B.Tech	9747474128	GJ06FH4563	QR View Del
PRIT Electrical · 25SC02022	B.Tech	9974151674	GJ06AN3245	QR View Del

➤➤ **Thank You..**